

Molecule Architect AI

"Instead of searching for molecules, AI invents them."

Main Idea

Scientists usually spend years testing thousands of molecules before finding one with the required properties. AI can reverse the process. Instead of asking "Here is a molecule. What does it do?", we ask "What material do we want?" and the AI proposes a conceptual molecular design. This is called inverse molecular design.

Objective

Build a free educational demo where the user enters a desired property (e.g., "Design a plastic that dissolves naturally in seawater."). The AI returns: Molecular backbone Functional groups Bond types Predicted properties Possible applications Limitations Always display: **"This molecule is AI-generated and hypothetical. It has NOT been experimentally verified."**

Materials

Laptop, Internet, Google AI Studio (free), HTML, CSS, JavaScript, optional poster and simple SVG molecule drawing.

Architecture

User → Desired Property → Prompt Engineering → Gemini/GPT → Conceptual Molecule Design → Display Results.

Prompt

You are an educational molecular design assistant. When given a desired property, create a conceptual molecular proposal. Return: 1. Molecular backbone 2. Functional groups 3. Bond types 4. Predicted physical properties 5. Possible applications 6. Scientific explanation 7. Limitations 8. State clearly that this is hypothetical and not experimentally verified.

Poster

Problem → AI Solution → Advantages (Faster research, Lower cost, Less waste) → Disclaimer.

5-Minute Presentation

1. Problem (45 s) 2. AI Solution (45 s) 3. Live Demo (2 min) 4. Real-world Applications (1 min) 5. Conclusion (30 s): AI helps scientists discover promising molecules faster; experiments are still essential.

Judging Strength

Choice of Topic: 10/10

Presentation: 9–10/10

Innovation: 10/10

Clarity: 9–10/10

Overall Impact: 10/10